

Repeated Games

Consider the game of *Iterated Prisoner's Dilemma*, in which players repeatedly engage in the Prisoner's Dilemma (the '*constituent game*').

Suppose the prisoners encounter for T periods.

What are

- Nash equilibria?
- Subgame perfect equilibria?

		(C, C)	$(3, 3)$
		(D, C)	$(5, 0)$
		(D, C)	$(5, 0)$
$\vdots$		$\vdots$	
		(D, D)	$(1, 1)$

Iterated Prisoner's Dilemma

Suppose the prisoners encounter for T periods.

- Are Nash equilibria 'good?'
- Are Subgame perfect equilibria 'good?'
- What should be the best for both?

		(C, C)	$(3, 3)$
		(D, C)	$(5, 0)$
		(D, C)	$(5, 0)$
$\vdots$	$\vdots$		
		(D, D)	$(1, 1)$

Iterated Prisoner's Dilemma

		(C, C)	$(3, 3)$
		(D, C)	$(5, 0)$
		(D, C)	$(5, 0)$
$\vdots$	$\vdots$		
		(D, D)	$(1, 1)$

What if the game is infinitely repeated?

Repeated Games

DEFINITION. Let $G = \langle N, (A_i), (\succeq_i) \rangle$ be a strategic game; let $A = \times_{i \in N} A_i$. An **infinitely repeated game** of G is an extensive game with perfect information and simultaneous moves $\langle N, H, P, (\succeq_i^*) \rangle$ in which

- $H = \{\emptyset\} \cup (\cup_{t=1}^{\infty} A^t) \cup A^{\infty}$ (where $\emptyset$ is the initial history and A^{∞} is the set of infinite sequences $(a^t)_{t=1}^{\infty}$ of action profiles in G)

Note:

$$H = \{\emptyset\} \cup (\cup_{t=1}^{\infty} A^t) \cup A^{\infty}$$

A history is a sequence of outcomes (action profiles),
of the constituent game.

- $\emptyset$ is the initial history
- A^t is the set of histories of length t . A history $(a^t) = (a^1, a^2, \dots, a^t) \in A^t$ is a sequence of outcomes: a^i is the outcome of period i .
- A^{∞} is the set of infinite sequences (*'terminal history'*) $(a^t)_{t=1}^{\infty}$ of action profiles.

- $P(h) = N$ for each nonterminal history $h \in H$.
- $\succeq_i^*$ is a preference relation on A^∞ that extends the preference relation $\succeq_i$ in the constituent game, in the sense that it satisfies the following condition of *weak separability*: if $(a^1, a^2, a^3, \dots) \in A^\infty$ is a terminal history, $a \in A$ and $a' \in A$ are action profiles of a constituent game, and $a \succeq_i a'$ then

$$(a^1, \dots, a^{t-1}, a, a^{t+1}, \dots) \succeq_i^* (a^1, \dots, a^{t-1}, a', a^{t+1}, \dots)$$
 for all values of t .

We further assume a payoff function u_i :

$$(a^t) \succeq_i^* (b^t)$$

iff

$$u_i(a^t) \geq u_i(b^t).$$

Preference Relations

$$\text{If } a \succeq_i a' \text{ then} \\ (a^1, \dots, a^{t-1}, a, a^{t+1}, \dots) \succeq_i^* (a^1, \dots, a^{t-1}, a', a^{t+1}, \dots).$$

The preference relation $\succeq_i^*$ is too general!

We consider three forms of preference relations:

- Discounting
- Limit of Means
- Overtaking

Discounting Preference Relations

EXAMPLE. Consider two sequences of payoffs of player i :

$(v_i^t) = (v_i^1, v_i^2, v_i^3 \dots)$ is evaluated by $\sum_{t=1}^{\infty} \delta^{t-1} v_i^t$

$(w_i^t) = (w_i^1, w_i^2, w_i^3 \dots)$ is evaluated by $\sum_{t=1}^{\infty} \delta^{t-1} w_i^t$

There is some number $\delta \in (0,1)$ (the *discount factor*) such that the sequence (v_i^t) of real numbers is at least as good as the sequence (w_i^t) if and only if $\sum_{t=1}^{\infty} \delta^{t-1} (v_i^t - w_i^t) \geq 0$.

Discounting Preference Relations

EXAMPLE.

$$(1, 1, 1, 1, \underbrace{0, 0, 0, 0, \dots}_{\text{all } 0})$$

$$(0, 0, 0, 0, \underbrace{2, 2, 2, 2, \dots}_{\text{all } 2})$$

If $\delta = \frac{1}{2}$, then they are evaluated by

$$1 + 1 \times \frac{1}{2} + 1 \times \frac{1}{4} + 1 \times \frac{1}{8} + 0 \times \frac{1}{16} + 0 \times \frac{1}{32} + \dots = 1\frac{7}{8}$$

$$0 + 0 \times \frac{1}{2} + 0 \times \frac{1}{4} + 0 \times \frac{1}{8} + 2 \times \frac{1}{16} + 2 \times \frac{1}{32} + \dots = \frac{1}{4}$$

So the first one is 'better.'

Limit of Means Preference Relations

EXAMPLE. Consider two sequences of payoffs:

$(v_i^t) = (v_i^1, v_i^2, v_i^3 \dots)$ is evaluated by $\lim_{T \rightarrow \infty} \sum_{t=1}^T \frac{v_i^t}{T}$

$(w_i^t) = (w_i^1, w_i^2, w_i^3 \dots)$ is evaluated by $\lim_{T \rightarrow \infty} \sum_{t=1}^T \frac{w_i^t}{T}$

The sequence (v_i^t) of real numbers is preferred to the sequence (w_i^t) iff $\liminf \sum_{t=1}^T \frac{(v_i^t - w_i^t)}{T} > 0$, (i.e., there exists $\varepsilon > 0$ such that $\sum_{t=1}^T \frac{(v_i^t - w_i^t)}{T} > \varepsilon$ for all but a finite number of periods T).

MATHEMATICAL NOTATION

The *infimum* $\inf S$ of a set S of real numbers is a number m such that $m \leq x$ for all $x \in S$, but for any positive ε there always exists $x' \in S$ such that $x' < m + \varepsilon$.

EXAMPLES (<http://en.wikipedia.org/wiki/Infimum>)

$$\inf\{x \in \mathbb{R}: 0 < x < 1\} = 0$$

$$\inf\{x \in \mathbb{R}: x^3 > 2\} = 2^{\frac{1}{3}}$$

$$\inf\{(-1)^n + \frac{1}{n} : n = 1, 2, 3, \dots\} = -1$$

MATHEMATICAL NOTATION

(http://en.wikipedia.org/wiki/Infimum_limit)

The *infimum limit* (or *limit inferior*) of a sequence $(a_i)_{i=1}^{\infty}$ of real numbers is

$$\liminf(a_i)_{i=1}^{\infty} = \lim_{n \rightarrow \infty} \inf\{a_k : k \geq n\}$$

Note that $\liminf(a_i)_{i=1}^{\infty}$ can be written as $\liminf a_i$.

Q: What is meant by $\liminf a_i > 0$?

A: It means that there exists $\varepsilon > 0$ such that $a_i > \varepsilon$ for all but some (finitely many) i . (We must eventually have $a_i > \varepsilon$ when i is large enough.)

Limit of Means Preference Relations

EXAMPLE. Consider two sequences of payoffs:

$(v_i^t) = (v_i^1, v_i^2, v_i^3 \dots)$ is evaluated by $\lim_{T \rightarrow \infty} \sum_{t=1}^T \frac{v_i^t}{T}$

$(w_i^t) = (w_i^1, w_i^2, w_i^3 \dots)$ is evaluated by $\lim_{T \rightarrow \infty} \sum_{t=1}^T \frac{w_i^t}{T}$

The sequence (v_i^t) of real numbers is preferred to the sequence (w_i^t) iff $\liminf \sum_{t=1}^T \frac{(v_i^t - w_i^t)}{T} > 0$, (i.e., there exists $\varepsilon > 0$ such that $\sum_{t=1}^T \frac{(v_i^t - w_i^t)}{T} > \varepsilon$ for all but a finite number of periods T).

EXAMPLE. Consider the following two sequences

$$\begin{array}{c} (\underbrace{0, \dots, 0}_{\substack{\text{finite no.} \\ \text{of zeros}}} , \underbrace{2, 2, \dots}_{\text{all } 2}) \\ (\underbrace{1, 1, 1, \dots}_{\text{all } 1}) \end{array}$$

- Which one is more preferred according to discounting preferences?
- Which one is more preferred according to the limit-of-means?

Overtaking Preference Relations

EXAMPLE. Consider two sequences of payoffs:

$(v_i^t) = (v_i^1, v_i^2, v_i^3 \dots)$ is evaluated by $\lim_{T \rightarrow \infty} \sum_{t=1}^T v_i^t$

$(w_i^t) = (w_i^1, w_i^2, w_i^3 \dots)$ is evaluated by $\lim_{T \rightarrow \infty} \sum_{t=1}^T w_i^t$

The sequence (v_i^t) is preferred to the sequence (w_i^t) if and only if $\liminf \sum_{t=1}^T (v_i^t - w_i^t) > 0$, *i.e.*, there exists $\varepsilon > 0$ such that $\sum_{t=1}^T (v_i^t - w_i^t) > \varepsilon$ for all but a finite number of periods T .

EXAMPLES.

- By the discounting criterion, which of $(1, -1, \underbrace{0, 0, \dots}_{\text{all } 0})$ and $(\underbrace{0, 0, \dots}_{\text{all } 0})$ is more preferred with $\delta = _?$
- By the limit of means criterion, which of $(1, -1, \underbrace{0, 0, \dots}_{\text{all } 0})$ and $(\underbrace{0, 0, \dots}_{\text{all } 0})$ is more preferred?
- By the overtaking criterion, which of $(1, -1, \underbrace{0, 0, \dots}_{\text{all } 0})$ and $(\underbrace{0, 0, \dots}_{\text{all } 0})$ is more preferred?

EXAMPLES.

- By the discounting criterion, which of $(-1, 2, \underbrace{0, 0, \dots}_{\text{all } 0})$ and $(\underbrace{0, 0, \dots}_{\text{all } 0})$ is more preferred with $\delta = _?$
- By the limit of means criterion, which of $(-1, 2, \underbrace{0, 0, \dots}_{\text{all } 0})$ and $(\underbrace{0, 0, \dots}_{\text{all } 0})$ is more preferred?
- By the overtaking criterion, which of $(-1, 2, \underbrace{0, 0, \dots}_{\text{all } 0})$ and $(\underbrace{0, 0, \dots}_{\text{all } 0})$ is more preferred?

EXAMPLE.

- By the discounting criterion, which of $(\underbrace{0, \dots, 0}_{M \text{ zeros}}, \underbrace{1, 1, \dots}_{\text{all 1}})$ and $(1, \underbrace{0, 0, \dots}_{\text{all 0}})$ is more preferred with $\delta = _?$
- By the limit-of-means criterion, which of $(\underbrace{0, \dots, 0}_{M \text{ zeros}}, \underbrace{1, 1, \dots}_{\text{all 1}})$ and $(1, \underbrace{0, 0, \dots}_{\text{all 0}})$ is more preferred?
- By the overtaking criterion, which of $(\underbrace{0, \dots, 0}_{M \text{ zeros}}, \underbrace{1, 1, \dots}_{\text{all 1}})$ and $(1, \underbrace{0, 0, \dots}_{\text{all 0}})$ is more preferred?

δ -Discounted Infinitely Repeated Games

Given a strategic game $G = \langle N, (A_i), (u_i) \rangle$, the **δ -Discounted Infinitely Repeated Game** of G is the infinitely repeated game for which the constituent game is G and the preference ordering $\succeq_i^*$ of each player $i \in N$ is derived from u_i using the discounting criterion with a discount factor of δ for each player.

Limit of Means Infinitely Repeated Games

Given a strategic game $G = \langle N, (A_i), (u_i) \rangle$, the **Limit of Means Infinitely Repeated Game** of G is the infinitely repeated game for which the constituent game is G and the preference ordering $\succeq_i^*$ of each player $i \in N$ is derived from u_i using the limit of means criterion for each player.

Overtaking Infinitely Repeated Games

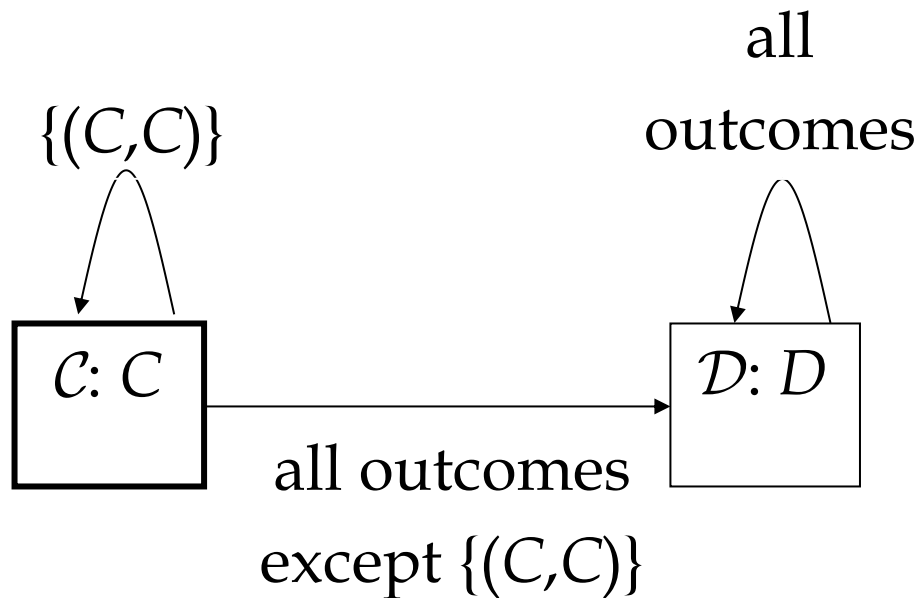
Given a strategic game $G = \langle N, (A_i), (u_i) \rangle$, the **Overtaking Infinitely Repeated Game** of G is the infinitely repeated game for which the constituent game is G and the preference ordering $\succeq_i^*$ of each player $i \in N$ is derived from u_i using the overtaking criterion for each player.

Strategies as Machines

A strategy of player i in an infinitely repeated game is a function that assigns an action in A_i to every finite sequence of outcomes in G .

Strategies as Machines

EXAMPLE. A 'grim' strategy for Iterated Prisoner's Dilemma.

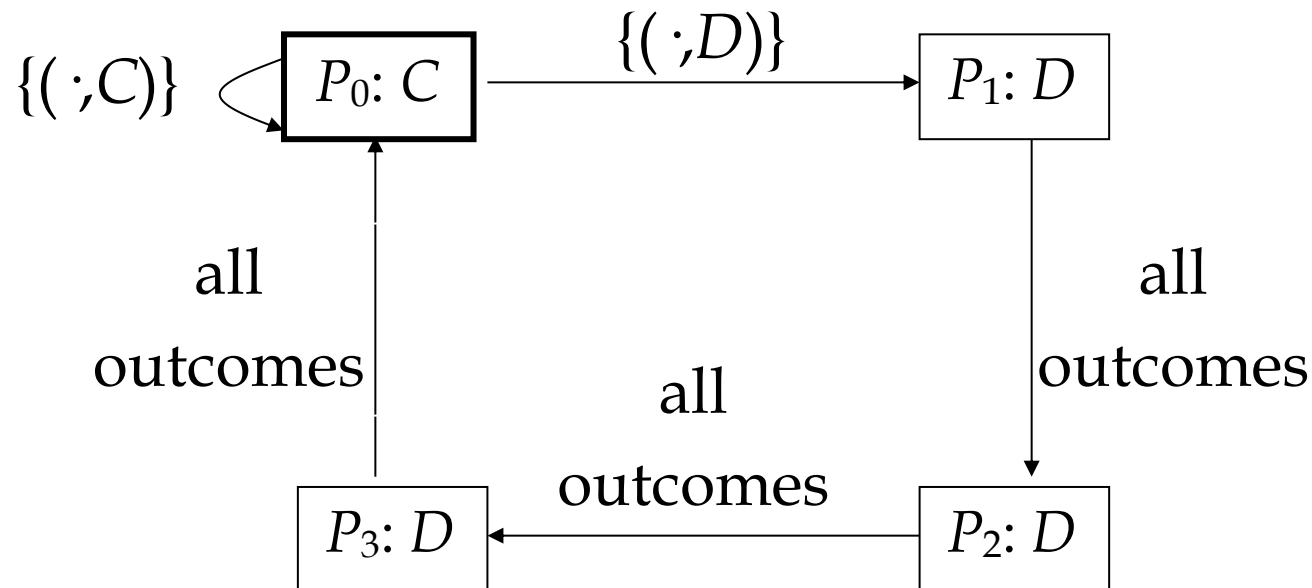


The FA

- $Q_i = \{\mathcal{C}, \mathcal{D}\}$
- $q_i^0 = \mathcal{C}$
- $f_i(\mathcal{C}) = C, f_i(\mathcal{D}) = D$
- $\tau_i(\mathcal{C}, (C, C)) = \mathcal{C}$
- $\tau_i(\mathcal{X}, (Y, Z)) = \mathcal{D}$ if $(\mathcal{X}, (Y, Z)) \neq (\mathcal{C}, (C, C))$

Strategies as Machines

EXAMPLE. A 'punishing-forgiving' strategy for Iterated Prisoner's Dilemma.



Strategies in Infinitely Repeated Games as Finite Automata

Not every strategy in an infinitely repeated game can be executed by a machine with a *finite* number of states.

Nash Equilibrium

What are the
Nash equilibria
of Infinitely Repeated Games?

Infinitely Repeated Games

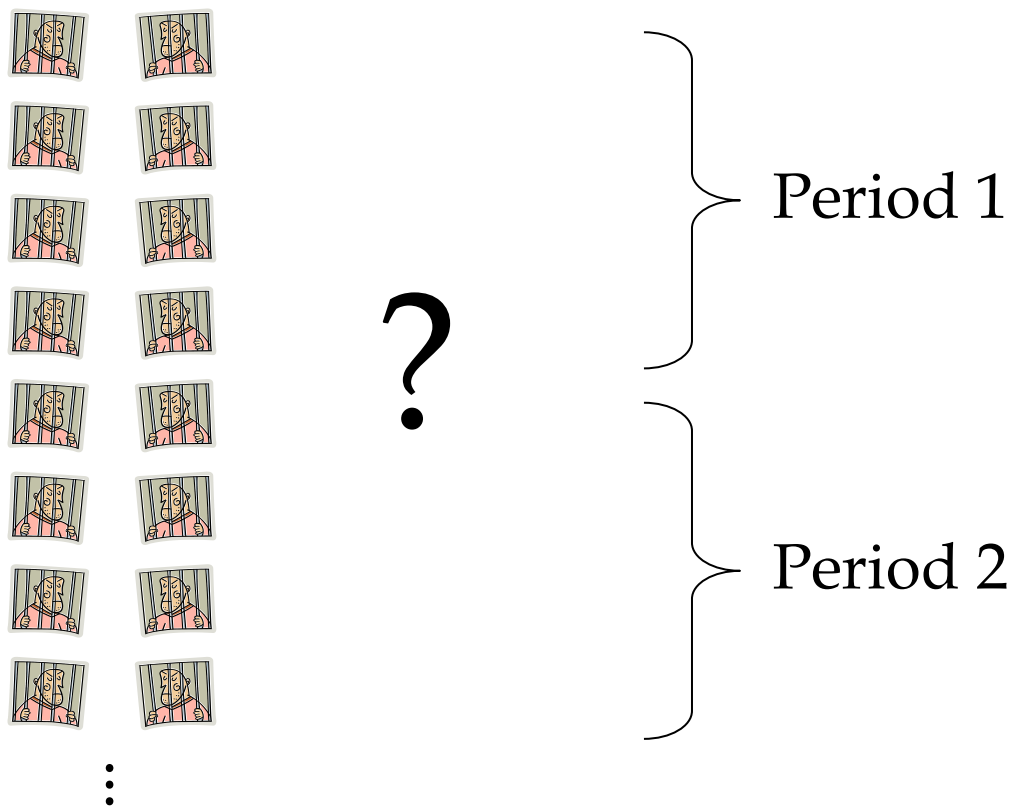
		(C, D)	$(0, 5)$	} Period 1
		(C, C)	$(3, 3)$	
		(D, C)	$(5, 0)$	
		(D, C)	$(5, 0)$	
		(C, D)	$(0, 5)$	} Period 2
		(C, C)	$(3, 3)$	
		(D, C)	$(5, 0)$	
		(D, C)	$(5, 0)$	
$\vdots$	$\vdots$			

Infinitely Repeated Games

		(C, D)	$(0, 5)$	} Period 1
		(C, C)	$(3, 3)$	
		(D, C)	$(5, 0)$	
		(D, C)	$(5, 0)$	
		(C, D)	$(0, 5)$	} Period 2
		(C, C)	$(3, 3)$	
		(D, C)	$(5, 0)$	
		(D, C)	$(5, 0)$	
$\vdots$	$\vdots$	$\vdots$	$\vdots$	

Limit of Means Payoff Profile: $(\frac{13}{4}, 2)$

Infinitely Repeated Games



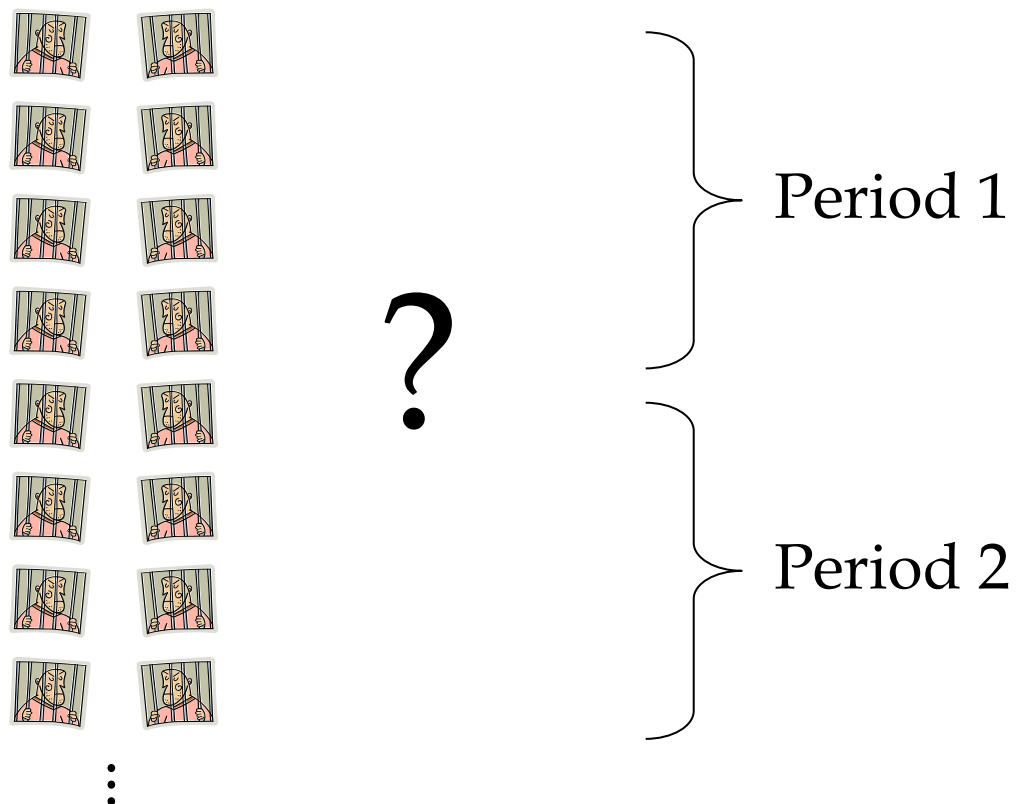
Limit of Means Payoff Profile: $(\frac{1}{4}, 4)$

Infinitely Repeated Games

		(D, D)	$(1, 1)$	} Period 1
		(C, D)	$(0, 5)$	
		(C, D)	$(0, 5)$	
		(C, D)	$(0, 5)$	
		(D, D)	$(1, 1)$	} Period 2
		(C, D)	$(0, 5)$	
		(C, D)	$(0, 5)$	
		(C, D)	$(0, 5)$	
$\vdots$	$\vdots$	$\vdots$	$\vdots$	

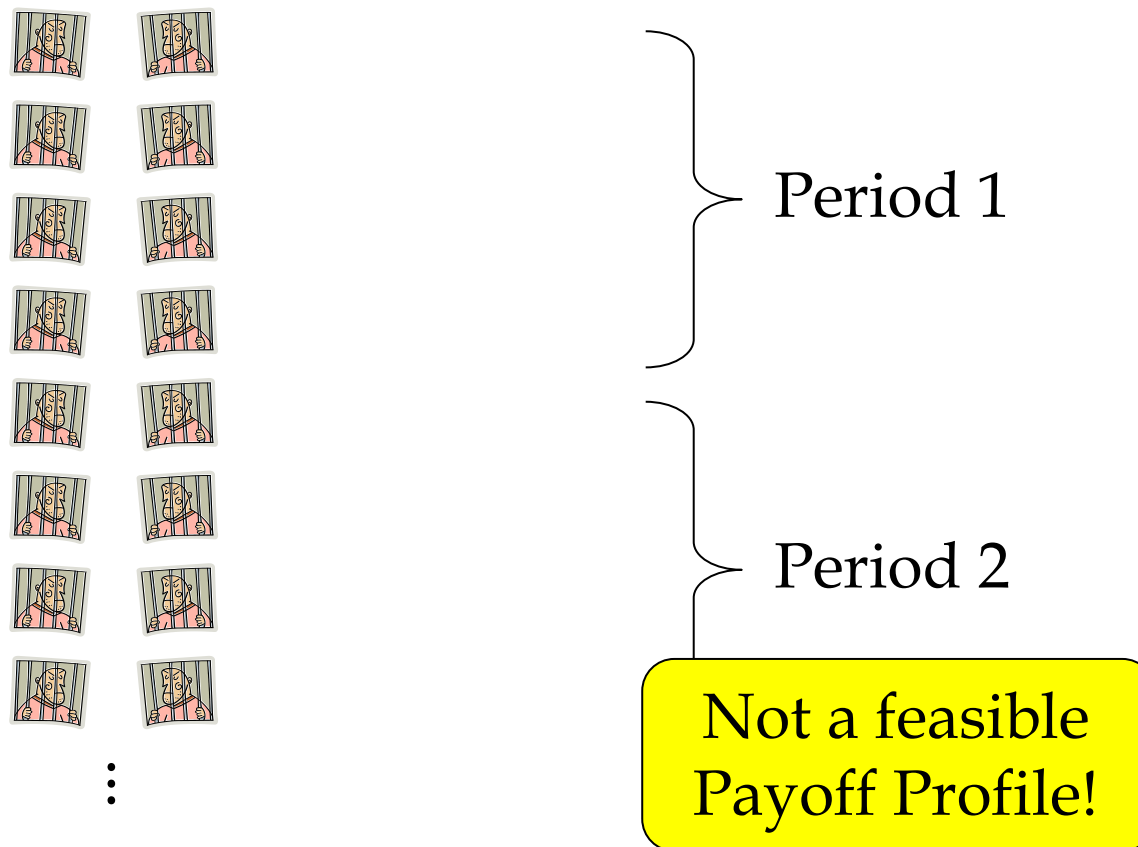
Limit of Means Payoff Profile: $(\frac{1}{4}, 4)$

Infinitely Repeated Games



Limit of Means Payoff Profile: $(0,3)$

Infinitely Repeated Games

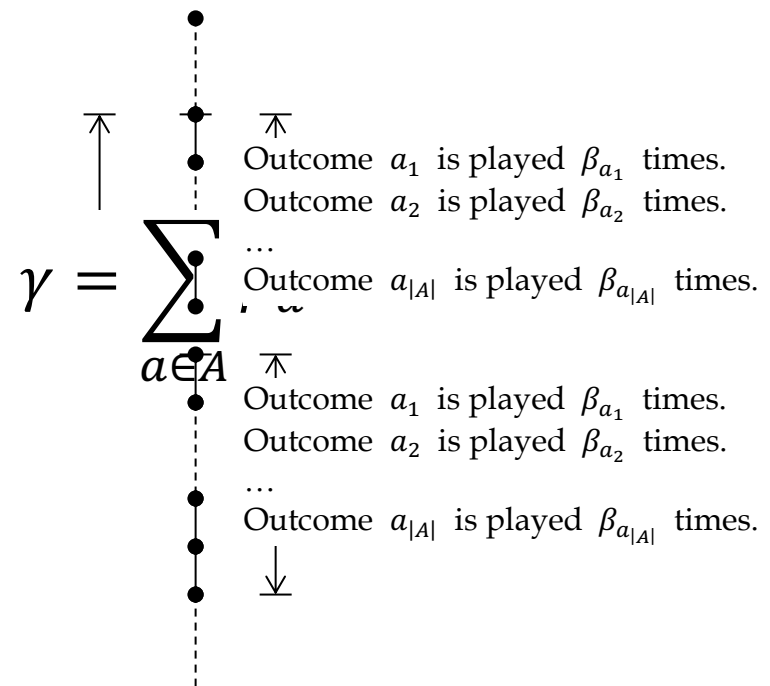


Limit of Means Payoff Profile: $(0,3)$

Infinitely Repeated Games

Let $G = \langle N, (A_i), (\succeq_i) \rangle$. The set of outcomes is $A = \times_{i \in N} A_i$.

Consider the players play an indefinitely repetition of a cycle of outcomes of length γ , in which each outcome $a \in A$ is played β_a times.



Infinitely Repeated Games

If the limit of means criterion is used, the payoff of player i in each period, **and hence the entire game**, is

$$w_i = \sum_{a \in A} \frac{\beta_a \cdot u_i(a)}{\gamma}.$$

$$\gamma = \sum_{a \in A} \beta_a$$

↑

⌋
Outcome a_1 is played β_{a_1} times.
Outcome a_2 is played β_{a_2} times.
...
Outcome a_m is played β_{a_m} times.
⌋

⌋
Outcome a_1 is played β_{a_1} times.
Outcome a_2 is played β_{a_2} times.
...
Outcome a_m is played β_{a_m} times.
↓

Payoff Profile (n users)

$$w = \left(\underbrace{\sum_{a \in A} \frac{\beta_a \cdot u_1(a)}{\gamma}}_{w_1}, \underbrace{\sum_{a \in A} \frac{\beta_a \cdot u_2(a)}{\gamma}}_{w_2}, \dots, \underbrace{\sum_{a \in A} \frac{\beta_a \cdot u_n(a)}{\gamma}}_{w_n} \right)$$

Feasible Payoff Profiles

$$\begin{aligned} w &= (w_1, w_2, \dots, w_n) \\ &= \left(\underbrace{\sum_{a \in A} \frac{\beta_a \cdot u_1(a)}{\gamma}}_{w_1}, \underbrace{\sum_{a \in A} \frac{\beta_a \cdot u_2(a)}{\gamma}}_{w_2}, \dots, \underbrace{\sum_{a \in A} \frac{\beta_a \cdot u_n(a)}{\gamma}}_{w_n} \right) \end{aligned}$$

A vector $w = \sum_{a \in A} \frac{\beta_a}{\gamma} u(a)$, where $\gamma = \sum_{a \in A} \beta_a$ and $\beta_a \geq 0$ for all a , is called a **feasible payoff profile** of $G = \langle N, (A_i), (u_i) \rangle$. Note that a feasible payoff profile is not necessarily a payoff profile of $G = \langle N, (A_i), (u_i) \rangle$.

Feasible Payoff Profiles

		(C, D)	(0, 5)	(D, D)	(1, 1)	(D, D)	(1, 1)
		(C, C)	(3, 3)	(C, C)	(3, 3)	(C, D)	(0, 5)
		(D, C)	(5, 0)	(D, D)	(1, 1)	(C, D)	(0, 5)
		(D, C)	(5, 0)	(C, D)	(0, 5)	(C, D)	(0, 5)
		(C, D)	(0, 5)	(D, D)	(1, 1)	(D, D)	(1, 1)
		(C, C)	(3, 3)	(C, C)	(3, 3)	(C, D)	(0, 5)
		(D, C)	(5, 0)	(D, D)	(1, 1)	(C, D)	(0, 5)
		(D, C)	(5, 0)	(C, D)	(0, 5)	(C, D)	(0, 5)
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
		$\left(\frac{13}{4}, 2\right)$			$\left(\frac{5}{4}, \frac{10}{4}\right)$		$\left(\frac{1}{4}, 4\right)$

Not a
feasible
Payoff
Profile!

⋮ ⋮ ⋮
(0,3)

Feasible Payoff Profiles

$$w = (w_i)_{i \in N} = \left(\sum_{a \in A} \frac{\beta_a \cdot u_i(a)}{\gamma} \right)_{i \in N} = \sum_{a \in A} \frac{\beta_a \cdot u(a)}{\gamma}$$

But how can we enforce this feasible payoff profile?

The feasible payoff profile is enforceable if no player can increase his payoff by deviating!

$$\gamma = \sum_{a \in A} \left(\begin{array}{l} \text{Outcome } a_1 \text{ is played } \beta_{a_1} \text{ times.} \\ \text{Outcome } a_2 \text{ is played } \beta_{a_2} \text{ times.} \\ \dots \\ \text{Outcome } a_m \text{ is played } \beta_{a_m} \text{ times.} \end{array} \right)$$

Class Discussion.

		(D, D)	$(1, 1)$	} Period 1
		(C, D)	$(0, 5)$	
		(C, D)	$(0, 5)$	
		(C, D)	$(0, 5)$	
		(D, D)	$(1, 1)$	} Period 2
		(C, D)	$(0, 5)$	
		(C, D)	$(0, 5)$	
		(C, D)	$(0, 5)$	
$\vdots$	$\vdots$	$\vdots$	$\vdots$	

Is this payoff profile enforceable? $(\frac{1}{4}, 4)$

Class Discussion.

		(C, C)	$(3, 3)$	} Period 1
		(C, C)	$(3, 3)$	
		(C, D)	$(0, 5)$	
		(C, D)	$(0, 5)$	
		(C, C)	$(3, 3)$	} Period 2
		(C, C)	$(3, 3)$	
		(C, D)	$(0, 5)$	
		(C, D)	$(0, 5)$	
$\vdots$	$\vdots$			

Is this payoff profile enforceable? $(\frac{3}{2}, 4)$

Enforceable Payoffs

In iterated prisoner's dilemma with limit of means preferences, payoff 1 is the worst possible payoff any player can get. Hence,

- the payoff profile $(\frac{1}{4}, 4)$ *is not enforceable* because player 1 gets a payoff of less than 1 — player 1 can always deviate and get 1.
- the payoff profile $(\frac{3}{2}, 4)$ *is enforceable* because every player gets a payoff of more than 1. (*Reason follows soon...*)

Enforceability of Payoffs and Outcomes of Constituent Game

Consider the constituent game G .

Player i 's **minmax** payoff:

$$v_i = \min_{a_{-i} \in A_{-i}} \max_{a_i \in A_i} u_i(a_{-i}, a_i)$$

The collection of actions

$$p_{-i} = \arg \min_{a_{-i} \in A_{-i}} \max_{a_i \in A_i} u_i(a_{-i}, a_i)$$

is the most severe 'punishment' that other players can inflict upon player i in the constituent game G .

Enforceability of Payoffs and Outcomes of Constituent Game

Player i 's minmax payoff:

$$v_i = \min_{a_{-i} \in A_{-i}} \max_{a_i \in A_i} u_i(a_{-i}, a_i).$$

- A payoff profile w is **enforceable** if and only if $w_i \geq v_i$ for all $i \in N$; it is **strictly enforceable** if and only if $w_i > v_i$ for all $i \in N$.
- An outcome $a \in A$ is a **(strictly) enforceable outcome** of G , if $u(a)$ is (strictly) enforceable in G .

Class Discussion

EXAMPLE. Consider the Iterated Prisoner's Dilemma with the limit of means criterion.

- What is the player 1's minmax payoff v_1 ?
- What is the player 2's minmax payoff v_2 ?
- Is the payoff profile $w = (\frac{1}{4}, 4)$ enforceable? Is it strictly enforceable?
- Is the payoff profile $w = (1, 1)$ enforceable? Is it strictly enforceable?

Class Discussion

EXAMPLE. Consider an infinitely repeated game of the game shown on the right.

- What is a player's minmax payoff v_i ?
- What is the corresponding 'most severe punishment' p_{-i} ?
- Name some feasible enforceable payoff profiles.
- Do you think player 1 will punish player 2?

	<i>A</i>	<i>D</i>
<i>A</i>	2,3	1,5
<i>D</i>	0,1	0,1

Nash Equilibrium of Limit of Means Infinitely Repeated Games

We consider a payoff profile w and $w_i < v_i$, then a player can change his strategy to increase his payoff to at least v_i in any period. Hence w cannot be a Nash equilibrium payoff profile.

Therefore, a Nash equilibrium payoff profile of a limit of means infinitely repeated game must be an enforceable payoff profile.

Nash Equilibrium of Limit of Means Infinitely Repeated Games

$(v_i^t) = (v_i^1, v_i^2, v_i^3 \dots)$ is evaluated by $\lim_{T \rightarrow \infty} \sum_{t=1}^T \frac{v_i^t}{T}$

PROPOSITION. Every Nash equilibrium payoff profile of the limit of means infinitely repeated game of $G = \langle N, (A_i), (u_i) \rangle$ is an enforceable payoff profile of G .

Trigger Strategies

Consider the following strategy:

- Always play according to the agreement, unless
- in a previous period, the other player deviated from the agreement. In this case, play D forever.

		(C, C)	$(3, 3)$
		(C, C)	$(3, 3)$
		(C, D)	$(0, 5)$
		(C, D)	$(0, 5)$
		(C, C)	$(3, 3)$
		(C, C)	$(3, 3)$
		(C, D)	$(0, 5)$
		(C, D)	$(0, 5)$
$\vdots$	$\vdots$	$\vdots$	$\vdots$
			$(\frac{3}{2}, 4)$

Trigger Strategies

Player i 's strategy s_i :

- In round t , player i always chooses a_i^t , unless
- in round t' in a previous period, player $j \neq i$ deviated and did not play $a_j^{t'}$. In this case, player i plays $(p_{-j})_i$, the most severe 'punishment' that can be inflicted upon player j by all players other than j .

		$(a_1^1, \dots, a_n^1)$
		$(a_1^2, \dots, a_n^2)$
		$\vdots$
		$(a_1^\gamma, \dots, a_n^\gamma)$
		$(a_1^1, \dots, a_n^1)$
		$(a_1^2, \dots, a_n^2)$
		$\vdots$
		$(a_1^\gamma, \dots, a_n^\gamma)$
$\vdots$		$\vdots$

Trigger Strategies

The strategy s_i is a trigger strategy.

The ‘normal’ payoff profile should be an enforceable payoff profile $w = \sum_{a \in A} \frac{\beta_a}{\gamma} u(a)$ (which is feasible).

If player j deviates, he receives his minmax payoff $v_j \leq w_j$ in every subsequent period (**Harsh!**).

What action should player j play when he is being punished?

Trigger Strategies

Therefore, the strategy profile s , in which all users are using a trigger strategy, is in Nash equilibrium.

Class Discussion

EXAMPLE. Consider the Iterated Prisoner's Dilemma with the limit of means criterion.

$$v = (\underbrace{1}_{v_1}, \underbrace{1}_{v_2})$$

- What is the corresponding 'most severe punishment' p_{-i} ?
- Name some feasible enforceable payoff profiles.
- Name some trigger strategies.

Nash Equilibrium of Limit of Means Infinitely Repeated Games

PROPOSITION. (*Nash folk theorem for the limit of means criterion*) Every feasible enforceable payoff profile of $G = \langle N, (A_i), (u_i) \rangle$ is a Nash equilibrium payoff profile of the limit of means infinitely repeated game of G .

Summary for Limit of Means Infinitely Repeated Games:

PROPOSITION. Every Nash equilibrium payoff profile of the limit of means infinitely repeated game of $G = \langle N, (A_i), (u_i) \rangle$ is an enforceable payoff profile of G .

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Do these results hold for δ -Discounted Infinitely
Repeated Games?

Nash Equilibrium of δ -Discounted Infinitely Repeated Games

We consider a payoff profile w and $w_i < v_i$, then a player can change his strategy to increase his payoff to at least v_i in any period. Hence w cannot be a Nash equilibrium payoff profile.

Therefore, a Nash equilibrium payoff profile of a δ -discounted infinitely repeated game must be an enforceable payoff profile.

Nash Equilibrium of δ -Discounted Infinitely Repeated Games

$$(v_i^t) = (v_i^1, v_i^2, v_i^3 \dots) \text{ is evaluated by } \sum_{t=1}^{\infty} \delta^{t-1} v_i^t$$

PROPOSITION. Every Nash equilibrium payoff profile of the δ -discounted infinitely repeated game of $G = \langle N, (A_i), (u_i) \rangle$ is an enforceable payoff profile of G , for any $\delta \in (0,1)$.

Are Trigger Strategies Effective in δ -Discounted Infinitely Repeated Games?

Player i 's strategy s_i :

- Player i always chooses a_i^t , unless
- in a previous period t' , player $j \neq i$ deviated and did not play $a_j^{t'}$. In this case, player i plays $(p_{-j})_i$ the most severe 'punishment' that can be inflicted upon player j by all players other than j .

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If δ is too small, then the deviant j cannot be 'punished!'

Nash Equilibrium of δ -Discounted Infinitely Repeated Games

PROPOSITION. (*Nash folk theorem for the discounting criterion*) Let w be a strictly enforceable feasible payoff profile of $G = \langle N, (A_i), (u_i) \rangle$. For all $\varepsilon > 0$ there exists $\underline{\delta} < 1$ such that if $\delta > \underline{\delta}$ then the δ -discounted infinitely repeated game of G has a Nash equilibrium whose payoff profile w' satisfies $|w' - w| < \varepsilon$.

Summary for δ -Discounted Infinitely Repeated Games:

PROPOSITION. Every Nash equilibrium payoff profile of the δ -discounted infinitely repeated game of $G = \langle N, (A_i), (u_i) \rangle$ is an enforceable payoff profile of G .

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